

The Quantum Random Oracle Toolbox

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Abstract. In which we reveal the tricks of the trade.

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1 Introduction

Hans: Idea: Take the PKE from “Random Oracles Are Practical” [BR93] as a motivating example running through the paper, using stronger and stronger techniques to show stronger and stronger security results.

There are three main challenges for proofs in the quantum random oracle model, relating to the application of standard techniques in the classical random oracle model:

1. Recording: or, how to extract QRO queries.
2. Reprogramming: or, how to alter specific outputs of the QRO.
3. Rewinding: or, how to derandomize quantum oracle algorithms.

2 Preliminaries

Basic notation. We take the natural numbers $\mathbb{N}$ to include 0. For $n \in \mathbb{N}$, we define the set $[n] = \{0, \dots, n-1\}$. We will employ the convention that $N = 2^n$; thus, $NM = 2^{n+m}$.

Functions are defined over bitstrings unless otherwise noted, and the set of all functions from the set of m -length bitstrings $\{0, 1\}^m$ to the set of n -length $\{0, 1\}^n$ bitstrings is denoted $\mathcal{F}_{m \rightarrow n}$.

We use $\mathbb{1}_{n \times n}$ to denote the n -by- n identity matrix, dropping the subscript whenever the dimension is clear from the context.

Hans: From SoK:

For a positive integer n , we let $[n]$ denote the set $\{1, \dots, n\}$. For a bit string $x \in \{0, 1\}^*$, $|x|$ denotes its length. For a finite set $\mathcal{X}$, $|\mathcal{X}|$ is the cardinality of $\mathcal{X}$. For a list $\mathbf{X}$, $\mathbf{X}[i]$ retrieves the i th element of the list, and, for an index set $\mathcal{I}$, $\mathbf{X}[\mathcal{I}]$ indicates the list $\mathbf{X}$ restricted to the elements whose indices are contained in $\mathcal{I}$.

3 A Brief Quantum Primer

Quantum theory arose as a branch of physics, and so much of the standard notational conventions are hand-wavy, and often inconsistently implemented across the literature. For the sake of clarity and self-containedness, we specify all our conventions below, providing a brief summary of the building blocks that make up quantum computing theory along the way; for more details, see the celebrated introductory text by Nielsen and Chuang [NC10].

Provide explanations of:

1. Bare-bones background on quantum states and Dirac notation; states vs. labels.
2. $N = 2^n$; the set $[n]$; n -length registers are described by vectors in an N -sized Hilbert space.
3. Greek letters: When and why I use them.
4. Precise notational conventions regarding registers.
5. Precise conventions regarding fonts.
6. Notation for unitary evolutions: $\xrightarrow{U}$.
7. Stick to pure states, and only mention the generalization to density matrices.
8. Basic background on game hopping and advantages.
9. Maybe write something about asymptotic vs. concrete security.

3.1 Dirac Notation

In general, quantum states are complex-valued vectors in a Hilbert space (a vector space defined over functions), which may or may not be finite-dimensional.¹ Quantum states are represented using the Dirac notation, in which a quantum state carrying the *label* ψ is written $|\psi\rangle$, its conjugate transpose is written $\langle\psi| := |\psi\rangle^\dagger := (|\psi\rangle^*)^T$, and its inner product with itself is written $\langle\psi|\psi\rangle := \|\psi\|^2$. Quantum states are unit-norm, and so we have that $\langle\psi|\psi\rangle = 1$. For two not necessarily equal quantum states $|\psi\rangle$ and $|\varphi\rangle$, their inner product $\langle\psi|\varphi\rangle$ is a complex number α with norm $|\alpha|$ between zero and one, where “zero”

¹ For example, consider the state $|p(x)\rangle$ describing the position $x \in [a, b]$ of particle p ; this will live in an infinite-dimensional Hilbert space characterized by the basis states $\{|p(x)\rangle\}_{x \in [a, b]}$.

means the states are orthogonal, and “one” means the states are identical (modulo an unobservable global phase factor $e^{i\theta}$).

This calculus is fine, but one may wonder, what *is* a quantum state—in the sense, what kind of mathematical object is it? The answer can be found in the statement that they live in N -dimensional *Hilbert space*: Hilbert spaces are vector spaces defined over *functions* in place of vectors, where the inner product between two functions f and g on the interval $[a, b]$ is defined to be $\int_{-\infty}^{\infty} f(x)^* g(x) dx$. This inner product defines a norm, and a notion of orthogonality, from which a vector space may be constructed. Quantum states, then, are the functions that arise as solutions to a differential equation describing the dynamics of the system, i.e. the Schrödinger equation in the non-relativistic, non-spin case; the Pauli equation in the case with spin (such as electrons), and the Dirac equation in the relativistic case (excluding gravity). They take some input variable corresponding to an observable, such as the position x , and returns the *amplitude* corresponding to that position; squaring this then gives the probability that observing the particle will yield x as a result.

If all of this sounds terribly complicated, then you understand why physicists grew so fond of Dirac’s notation, which neatly swept all of these horrible details under a rug of conceptual clarity!

Thankfully, things become a bit easier in the discrete case: the Hilbert space is now finite, and so it can be expressed in terms of a finite set of basis “vectors” (i.e., functions), and the inner product is now a sum rather than an integral. Further, functions now have finite domains, and may thus be written as N -length vectors instead, where each element corresponds to an input to the function, enabling us to employ our toolbox from linear algebra to study them. This is the approach taken in the quantum information sciences.

3.2 Qubit Basis States

We are interested in finite-dimensional quantum states, implemented on a collection of n qubits. A *qubit* is a quantum state existing in the two-dimensional Hilbert space characterized by the basis states $|0\rangle$ and $|1\rangle$. In physical implementations, these states may for example be mapped to the spin of an electron (“up”/“down”), or the two lowest energy states of an atom, or in general any two orthogonal quantum states, where two quantum states are orthogonal if and only if there exists a measurement procedure that perfectly distinguishes them.

It is worth emphasizing that the “0” and “1” here are *labels*, and do not say anything about the physical system in question other than that it contains (at least) two orthogonal quantum states, which are being identified with the two basis states $|0\rangle$ and $|1\rangle$.

Since two qubit basis states $|b_1\rangle$ and $|b_2\rangle$, for $b_1, b_2 \in \{0, 1\}$, the two states must be either identical (up to the aforementioned phase factor) or orthogonal, we have that

$$\langle b_1 | b_2 \rangle = \delta_{b_1 b_2} ,$$

where δ_{ij} is the Kronecker delta, equal to 1 if $i = j$, and 0 otherwise.

3.3 Combined Quantum States

In general, two quantum systems (i.e. states describing distinct physical systems) may be combined via the tensor product (also known as the *outer product*):

$$|\psi\rangle \otimes |\varphi\rangle := |\psi\varphi\rangle .$$

If $|\psi\rangle$ describes an N -dimensional quantum system, and $|\varphi\rangle$ describes an M -dimensional quantum system, then the combined state $|\psi\varphi\rangle$ describing the combined system will be NM -dimensional. (Note how the dimensions are *multiplied* rather than added, as one might naïvely expect.)

Applying this to qubits, given a collection of n qubits $|b_i\rangle$, each initialized to one of the two basis states (so that $b_i \in \{0, 1\}$), we may construct the combined state

$$|b_0 b_1 \dots b_{n-1}\rangle := |b_0\rangle \otimes |b_1\rangle \otimes \dots \otimes |b_{n-1}\rangle .$$

Since each qubit is a two-dimensional quantum state, we see that the resulting combined state will have dimension $N := 2^n$. The resulting Hilbert space will be characterized by the computational² basis states

$$\{|i\rangle\}_{i \in [N]} = \{|b_0 b_1 \dots b_{n-1}\rangle\}_{b_0, \dots, b_{n-1} \in \{0, 1\}} .$$

² As with any vector space, Hilbert spaces permit an infinitude of basis choices; the “computational” basis is simply the name of the choice wherein basis states are labelled by classical bits (or, in general, bitstrings).

This exponentiality in the dimension of quantum states as the number of qubits increases sits at the core of the quantum advantage over classical computers. As an illustrative example, the Hilbert space corresponding to a three-qubit quantum system is characterized by the eight computational basis states:

$$\begin{aligned} \{|i\rangle\}_{i \in [2^3]} &= \{|000\rangle, |001\rangle, |010\rangle, |011\rangle, |100\rangle, |101\rangle, |110\rangle, |111\rangle\} \\ &= \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \right\}. \end{aligned}$$

As in the single-qubit case, we see that we have orthonormality over the computational basis states:

$$\langle x|y\rangle = \delta_{xy}.$$

3.4 Qubit Registers

Given a collection of n qubits, we will find it useful to divide it into *registers* of specified sizes, for example to separate distinct elements in an input. We will use subscripts to refer to specific registers, unless the register we are referring to is clear from context.

As an example, let's say we want the first n_1 qubits store the variable x —call this “Register 1”—and that the remaining n_2 qubits—“Register 2”—should store the variable y . Then the following are all equivalent ways of denoting the combined state:

$$|x||y\rangle = |x\rangle_1 \otimes |y\rangle_2 := |x\rangle_1 |y\rangle_2 := |x, y\rangle,$$

where $\parallel$ denotes string concatenation.

3.5 General Qubit States

Greek vs. Roman letters. We will use the letters $x, y, z, \dots$ to represent classical binary strings. Thus, a state $|x\rangle$ will always be understood to be a computational basis state. Whenever we are talking specifically about single-qubit computational basis states, we will use b to represent a single bit, and write $|b\rangle$.

General quantum states are traditionally labelled by letters taken from the middle of the greek alphabet, such as ψ, φ, ξ , etc. We will follow this tradition, or otherwise use a descriptive label relating to the specific context in which the state is being used. Quantum *amplitudes*, meanwhile, are written using letters taken from the start of the greek alphabet ($\alpha, \beta, \dots$).

Expanding a state in terms of the basis. A general single-qubit state may be written in terms of the computational basis states as follows:

$$|\psi\rangle = \alpha_0 |0\rangle + \alpha_1 |1\rangle,$$

where the amplitudes α_i are complex numbers satisfying $|\alpha_0|^2 + |\alpha_1|^2 = 1$, (where the absolute value of a complex number α is defined as $|\alpha| = \sqrt{\alpha^* \alpha}$).

Whenever a qubit is in a state described by amplitudes α_0 and α_1 that are both non-zero, we say that the qubit is in a *superposition* over the values “0” and “1”. If we then *observe* the qubit, the probability that the observed outcome X is either 0 or 1 is then given by $\Pr[X = b] = |\alpha_b|^2 = |\langle \psi | b \rangle|^2$, and after observation, the superposition will have “collapsed” to the basis state corresponding to the observed value, $|b\rangle$.³

³ The discernible effect of observations on quantum states has led to a century of heated philosophical debate over how quantum theory should be interpreted: The term “collapse” used here is typically associated with the conservative “Copenhagen interpretation”, championed by Niels Bohr, and commonly characterized as taking a “shut-up-and-calculate” attitude towards the problem. Another popular interpretation is the many-worlds interpretation due to Hugh Everett, and favoured by father of quantum computing theory David Deutsch, among others. Instead of the quantum state “collapsing” to become “classical”, this interpretation would say

Likewise, a general, n -qubit quantum state may be written in terms of the computational basis states:

$$|\psi\rangle = \sum_{\forall i \in [2^n]} \alpha_i |i\rangle = \sum_{i=0}^{N-1} \alpha_i |i\rangle = \alpha_0 |0 \dots 00\rangle + \alpha_1 |0 \dots 01\rangle + \dots + \alpha_{N-1} |1 \dots 11\rangle ,$$

where the *amplitudes* α_i again satisfy $\sum_{\forall i \in [2^n]} |\alpha_i|^2 = 1$.

When expanded over the basis states, the inner product of two quantum states $|\psi\rangle = \sum_{\forall i \in [N]} \alpha_i |i\rangle$ and $|\varphi\rangle = \sum_{\forall j \in [N]} \beta_j |j\rangle$ may be written

$$\langle \psi | \varphi \rangle = \sum_{\forall i, j \in [N]} \alpha_i^* \beta_j \langle i | j \rangle = \sum_{\forall i, j \in [N]} \alpha_i^* \beta_j \delta_{ij} = \sum_{\forall i \in [N]} \alpha_i^* \beta_i .$$

We see that we have recovered the inner product of complex-valued vectors, which is just the usual inner product of real-valued vectors with an extra complex conjugation.⁴ Thus we may apply the usual operational interpretation of the inner product, in which vectors are “perpendicular” when their inner product is 0, and “parallel” when it is 1. Together with the requirement that quantum states have unit norm, this means that an inner product of 1 implies that the two quantum states are actually identical, (up to the aforementioned unobservable phase factor).

When expanding quantum states in terms of basis states in the manner above, we will sometimes implicitly exclude terms for which $\alpha_i = 0$, writing instead

$$|\psi\rangle = \sum_i \alpha_i |x_i\rangle := \sum_{\forall i \in \mathcal{I}} \alpha_i |x_i\rangle ,$$

where $\mathcal{I} = [\ell]$ indexes the set $\{|x\rangle \mid x \in [N] \wedge \langle \psi | x \rangle \neq 0\}$ for some $\ell \in \mathbb{N}$.

3.6 Unitary Operations

Since we are working over finite-dimensional Hilbert spaces, quantum operations, i.e. evolutions of quantum states, may be represented as $N \times N$ -dimensional matrices. What properties should they have?

Clearly, any operation that we perform on a quantum state (*except* observing them!) should yield another valid quantum state as a result. Since quantum states must be unit-norm, this means that any operation we perform on them must preserve the norm. This fact turns out to be sufficient to restrict the space of allowed operations to the unitary matrices, i.e. invertible square matrices U satisfying $U^{-1} = U^\dagger$.

Exercise 1. Prove that requiring operations to be norm-preserving is equivalent to restricting the set of legal operations to the set of unitary matrices.

We will use capital roman letters to refer to quantum operations, in reference to their matrix representation. Whenever the qubit(s) that the operation is being applied to is not clear from context, we will provide the index of the qubit(s) in the subscript, e.g. U_i for single-qubit operations, and $U_{i,j}$ for two-qubit operations, where $i, j \in [N]$ and $i \neq j$ (or more generally, the *registers* on which the operations are applied).

It is often useful to refer to operations in terms of the effect they have on the starting state, and so we define $|\psi\rangle \xrightarrow{U} |\varphi\rangle$ to mean that we perform the operation $|\psi\rangle \longrightarrow U |\psi\rangle = |\varphi\rangle$.

Some elementary unitary transformations, known as quantum *gates*, include

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} , \quad Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} , \quad H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} , \quad CX = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} .$$

that both terms are ill-defined, and that the universe should instead be understood to be nothing but one big superposition state over all worlds that might have arisen from the initial conditions of the Big Bang; and that what actually happened when you made your observation, was that the quantum state describing the subsystem you were studying and the quantum state describing the subsystem containing *you* became correlated via the process known as *entanglement*. (In case you're wondering, I place my own bet on the *relational* interpretation due to Carlo Rovelli, which may be described as “many-worlds without the multiverse”.)

⁴ This is because inner products on any vector space must satisfy conjugate symmetry, $\langle x, y \rangle = \langle y, x \rangle^*$; see https://en.wikipedia.org/wiki/Inner_product_space.

Exercise 2. Write down “truth tables” for each of the above gates by writing down their effect on the (single-qubit resp. two-qubit) computational basis states. (E.g., $|0\rangle \xrightarrow{X} |1\rangle$, etc.)

The “C” in CX is short for “controlled”, where the value of the first qubit *controls* the operation on the second qubit. Any single-qubit operation U defines a two-qubit controlled version of the same operation, which we will refer to as CU .

Exercise 3. Starting with the two-qubit state $|\psi\rangle = |00\rangle$, apply the operations H_0 and $CX_{0,1}$. What are the possible outcomes of observing the resulting state, and what are their probabilities? Compare with the result of applying the single-qubit gates H_0 and H_1 instead.

Exercise 4. Write a new truth table for X, Z, and H, only now in terms of the alternative basis states $|+\rangle := \frac{|0\rangle+|1\rangle}{\sqrt{2}}$, $|-\rangle := \frac{|0\rangle-|1\rangle}{\sqrt{2}}$ instead. Compare with the truth table from Exercise 2. What has changed?

Exercise 5. Give the matrix representation of a unitary operation that acts on the alternative basis from Exercise 4 in the same way that CX acts on the computational basis.

Another two-qubit unitary gate that will turn out to be useful is the **SWAP** gate, defined as $|01\rangle \xrightarrow{\text{SWAP}} |10\rangle$, $|10\rangle \xrightarrow{\text{SWAP}} |01\rangle$, and acting like the identity on $|00\rangle$ and $|11\rangle$. Applying this to general qubit states, we get

$$\begin{aligned} (\alpha_0 |0\rangle + \alpha_1 |1\rangle) \otimes (\beta_0 |0\rangle + \beta_1 |1\rangle) &= \alpha_0 \beta_0 |00\rangle + \alpha_0 \beta_1 |01\rangle + \alpha_1 \beta_0 |10\rangle + \alpha_1 \beta_1 |11\rangle \\ \xrightarrow{\text{SWAP}} \alpha_0 \beta_0 |00\rangle + \alpha_0 \beta_1 |10\rangle + \alpha_1 \beta_0 |01\rangle + \alpha_1 \beta_1 |11\rangle &= (\beta_0 |0\rangle + \beta_1 |1\rangle) \otimes (\alpha_0 |0\rangle + \alpha_1 |1\rangle). \end{aligned}$$

Thus, we see that the **SWAP** gate has the effect of swapping the states of two qubits with each other.

Exercise 6. Give the matrix representation of the **SWAP** gate in the computational basis.

Transformations in space vs. time. You may have noticed that we are working with two different kinds of product when talking about unitary matrices: The matrix product $U_1 U_2$, and the tensor product $U_1 \otimes U_2$. When should each kind of product be used?

To illustrate, let $|\psi\rangle$ and $|\varphi\rangle$ be two single-qubit states, combined in the two-qubit state $|\psi, \varphi\rangle := |\psi\rangle \otimes |\varphi\rangle$. Then, if U_1 and U_2 are single-qubit transformations (i.e. 2×2 unitary matrices), and you want to apply U_1 to $|\psi\rangle$ and U_2 to $|\varphi\rangle$, you should write

$$U_{1,2} |\psi, \varphi\rangle := (U_1 \otimes U_2) (|\psi\rangle \otimes |\varphi\rangle).$$

The matrix $U_{1,2} = U_1 \otimes U_2$ will be a two-qubit transformation (i.e. a 4×4 unitary) that combine the effect of the two single-qubit transformations to apply them in parallel (that is, in the same time step).

Next, imagine instead that you have two two-qubit unitaries (4×4 matrices) U_1 and U_2 , and you want to apply U_1 first, and then U_2 . In other words, you want something to the effect of

$$|\psi(0), \varphi(0)\rangle \xrightarrow{U_1} |\psi(1), \varphi(1)\rangle \xrightarrow{U_2} |\psi(2), \varphi(2)\rangle.$$

This is then given by

$$U_2 U_1 |\psi(0), \varphi(0)\rangle = U_2 (U_1 |\psi(0), \varphi(0)\rangle) := U_2 |\psi(1), \varphi(1)\rangle := |\psi(2), \varphi(2)\rangle.$$

Note the reversed order of operations!

As before, we could define a unitary $U_{1,2} := U_1 U_2$. Thus, even though we want to say that the application of U_1 and U_2 happens in two time steps, we see that the time of application can be “collapsed” to a single time-step! Thus, when counting time steps in the analysis of quantum algorithms, the first step is to define some Turing-universal set of elementary quantum gates from which all other unitaries must be constructed, in series and in parallel.

To summarize, the tensor product is used when applying unitaries in *parallel*, while the matrix product is used when applying unitaries in *series*.

Exercise 7. Construct the **SWAP** gate as a series of CX gates.

3.7 Quantum Entities, Entanglement, and Towards a Quantum Pseudocode

So far, we have defined notation for computational basis states $|x\rangle$, general quantum states $|\psi\rangle$, unitary transformations U , and the result of applying a unitary transformation U on a quantum state $|\psi\rangle$, written $|\psi\rangle \rightarrow |\varphi\rangle = U|\psi\rangle$, or equivalently $|\psi\rangle \xrightarrow{U} |\varphi\rangle$.

And still this isn't enough. This is because we are going to be talking about communicating parties. Consider the situation that Alice has two quantum bits initialized to the zero state, so that her state is $|\psi\rangle = |0\rangle_0 |0\rangle_1$. Now she wants to send her second qubit register, $|0\rangle_1$, over to Bob, who is going to apply some single-qubit unitary to it, and then send it back. We might write this as $|0\rangle_1 \xrightarrow{U_1} |\xi\rangle_1$, where $|\xi\rangle_1$ is some unknown, single-qubit state, and say that Bob returns $|\xi\rangle_1$ to Alice at the end of the protocol.

But what if Alice had prepared her two qubits in the state $|\varphi\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$ instead? Then, the two qubits are *non-separable*, meaning that they can no longer be cleanly factored into a tensor product. In the world of quantum physics, this means that the two qubits are *entangled*—a fancy word describing a phenomenon with many deep consequences for physics and philosophy alike—but mathematically speaking it is nothing but the quantum generalization of correlated probabilities.⁵ Alice can still send her second qubit to Bob, and Bob can still apply the operation U to it; our problem is just that with the tools presented so far, we have no way of writing this operation without referencing the entire state, like so: $|\varphi\rangle \rightarrow |\xi\rangle = (\mathbb{1}_0 \otimes U_1)|\varphi\rangle$.⁶ What we want is to be able to write the operations of Alice and Bob without having to reference states belonging to the other, regardless of whether or not they share entanglement between them. In other words, we need a better *pseudocode*.

Let q be a physical qubit, in the sense of being some kind of particle or quantum system existing in the physical world. Now, guided by our intuition, we will say that Alice, instead of sending a quantum *state* to Bob, actually sends the *physical qubit* q .⁷ Anyone who receives a qubit, also gains the ability to apply any transformation or measurement on it, while those who sent it loses the ability to do so.

We will write the *state* of a qubit q as $|q\rangle$; so, for example, if q is in the state $|0\rangle$, then we write $|q\rangle = |0\rangle$. Additionally, it will sometimes be useful to reference how the state of q evolved over time. Let's say that at time t the unitary U was applied; this would then be written $|q(t)\rangle \xrightarrow{U} |q(t+1)\rangle$. Likewise, if q was initialized to $|0\rangle$, then $|q(0)\rangle = |0\rangle$.

Next, let us define an n -qubit quantum computer $\mathcal{Q}$ to be a list of qubits:

$$\mathcal{Q} := (q_0, \dots, q_{n-1}).$$

Then we may reference the i 'th qubit of $\mathcal{Q}$ by retrieving it from the list: $\mathcal{Q}[i] = q_i$. What's more, we may reference any collection of qubits by inputting a list of indices. This allows us to easily define arbitrary registers:

$$\mathcal{A} := \mathcal{Q}[\mathbf{I}] = (\mathcal{Q}[\mathbf{I}[0]], \mathcal{Q}[\mathbf{I}[1]], \dots, \mathcal{Q}[\mathbf{I}[\ell]]),$$

where $\ell = |\mathbf{I}|$ defines the size of the register $\mathcal{A}$.

In the case where there is no entanglement, we may write the state of a register $\mathcal{A}$ as

$$|\mathcal{A}\rangle := \bigotimes_{\forall i \in |\mathcal{A}|} |\mathcal{R}[i]\rangle = |q_{\mathbf{I}[0]}\rangle \otimes \dots \otimes |q_{\mathbf{I}[\ell]}\rangle = |q_{\mathbf{I}[0]} \dots q_{\mathbf{I}[\ell]}\rangle,$$

but of course, the entire point of this exercise was that we cannot write the whole state in terms of the individual qubit states in the general, entangled case, and so need an alternative way to reference its constituents.

⁵ In other words, the term “separable” here carries exactly the same meaning as the term “product” does when talking about classical probability distributions. More precisely, if an arbitrary quantum state is *separable* along some dividing line, such that $|\psi\rangle = |\varphi\rangle \otimes |\xi\rangle$, then the distribution over the observation outcomes will form a product distribution with the same dividing line. This remains true regardless of which basis you choose to make your observation in, implying that entanglement is an observation-independent (and therefore “objective”) property of the state.

⁶ For those ahead of the curve: One can reference one half of an entangled state using the generalized notion of a *mixed state*, by going to the density matrix formalization of quantum mechanics, which generalizes quantum states by combining them with classical probabilities. The reason why we avoid this solution here is that mixed states are not unique, and so there are many different states that could have led to Bob receiving a particular mixed state.

⁷ In real-world quantum technologies, qubits are typically stored in one kind of system (e.g. atoms) and sent using another kind (photons), but due to the non-cloneability of quantum states, together with the universal ability to transfer any quantum state from one qubit to another (known as the SWAP operation), our simplified model is without loss of generality.

Next, let Alice and Bob have access to quantum computers $\mathbb{A}$ and $\mathbb{B}$, respectively. Instead of saying that Alice sends a *state* to Bob, we will

Definition 1 (Qubit). A qubit q is a two-dimensional physical system, meaning that it is associated with an observable $\hat{q}$ taking one of two values, 0 or 1. A qubits are also associated with an address $i \in [n]$ (for some $n \in \mathbb{N}$).⁸ We will refer to the qubit with address i as q_i . If qubit q_i is in a pure state (i.e. not entangled with any other qubit), then its state may be denoted $|q_i\rangle$ (or, equivalently, $|q_i\rangle_i$). If $|q_i\rangle = |\psi\rangle$ for some state $|\psi\rangle$, then this may be denoted $|\psi\rangle_i$.

Definition 2 (Quantum memory). An n -qubit quantum memory is an n -element list of qubits $\mathcal{Q} = (q_1, \dots, q_n)$ comprising a closed quantum system, in the sense that no operation may be performed on the memory such that the system is no longer in a pure state⁹. In other words, the quantum memory can not entangle with any system outside of itself (though it may be observed, after which it immediately collapses to the observed basis state). Thus, the N -dimensional “global” quantum state $|\mathcal{Q}\rangle$ will always be well-defined.

Definition 3 (Quantum register). An m -qubit quantum register $\mathcal{A}$ is a subsystem of n -qubit quantum memory $\mathcal{Q}$, that is an ordered sub-list of $\mathcal{Q}$, where by ordered we mean that if $\mathcal{A} = (q_{i_0}, q_{i_1}, \dots, q_{i_m})$, then the qubit addresses $i_0, \dots, i_m \in [n]$ satisfy $i_0 < i_1 < \dots < i_m$. If the qubits in $\mathcal{A}$ are not entangled with any other subsystem of $\mathcal{Q}$, then the system $\mathcal{A}$ is in a pure state, which we may denote $|\mathcal{A}\rangle$. In a slight overload of notation, we let $[\mathcal{A}]$ denote the set $\{i_j\}_{j \in [m]}$ of qubit addresses associated with $\mathcal{A}$; then, if $|\mathcal{A}\rangle = |\psi\rangle$ for some specific state $|\psi\rangle$, this may be denoted $|\psi\rangle_{[\mathcal{A}]}$.

Definition 4 (Operating on registers). Let $\mathcal{Q}$ be an n -qubit quantum memory in some N -dimensional state $|\mathcal{Q}\rangle$, let $\mathcal{A}$ be an m -qubit register describing a subsystem of $\mathcal{Q}$, and let U be an $M \times M$ unitary matrix.

Let $U_{\mathcal{A}}$ be the $N \times N$ unitary matrix that has the effect of applying U on the qubits in $\mathcal{A}$ and the identity on the qubits in $\mathcal{Q} \setminus \mathcal{A}$:

$$U_{\mathcal{A}} := \text{SORT}^\dagger([\mathcal{A}], [\mathcal{Q} \setminus \mathcal{A}]) (U \otimes \mathbb{1}) \text{SORT}([\mathcal{A}], [\mathcal{Q} \setminus \mathcal{A}]),$$

where $\text{SORT}([\mathcal{A}], [\mathcal{Q} \setminus \mathcal{A}])$ is the $N \times N$ unitary matrix, composed of a series of SWAP gates, that has the effect of reordering the qubits $\mathcal{Q}$ such that the qubits of $\mathcal{A}$ precede the qubits of $\mathcal{Q} \setminus \mathcal{A}$; $\text{SORT}^\dagger$ reverses the operation of SORT ; and $\mathbb{1}$ is the $2^{n-m} \times 2^{n-m}$ identity matrix. Then, the application of U to the qubits in register $\mathcal{A}$ is captured in the method $\text{Apply}_{\mathcal{Q}}(U \rightarrow \mathcal{A})$, defined as

$$\text{Apply}_{\mathcal{Q}}(U \rightarrow \mathcal{A}) : |\mathcal{Q}\rangle \longrightarrow U_{\mathcal{A}} |\mathcal{Q}\rangle.$$

Whenever $\mathcal{Q}$ is clear from context, we will drop the subscript and simply write $\text{Apply}(U \rightarrow \mathcal{A})$.

⁸ In the formalism of quantum Turing machines given by Deutsch [Deu85], the addresses correspond to values of the *address observable* $\hat{x}$, except that we limit ourselves to finite quantum memory by letting the set of possible values be $[n]$ rather than $\mathbb{Z}$.

⁹ Note that this also implies that state preparation must be *deterministic*, or at least *posterior* to any randomness involved in the procedure: Flipping a (classical) coin and saying that you prepare a qubit in the state $|1\rangle$ on heads and $|0\rangle$ on tails would, when conditioning on the result of the coin flip, put $\mathcal{Q}$ in a mixed state, and must therefore be disallowed. Thankfully, due to Turing-universality, this restriction is without loss of generality.

Table 1. A summary of our notation.

Object	Notation	Example	Class/definition
Bitstring	Lowercase roman (end of alphabet)	x	$\in \{0, 1\}^n$
General state label	Lowercase greek (middle of alphabet)	ψ	$\in \{\text{labels}\}$
Qubit count	Lowercase roman (middle of alphabet)	n	$\in \mathbb{Z}$
Dimensionality	Uppercase roman (middle of alphabet)	N	$:= 2^n$
Amplitude	Lowercase greek (start of alphabet)	α	$\in \mathbb{C}$
Computational basis state	Dirac ket with bitstring label	$ x\rangle$	$\in \mathbb{C}^M, n \geq \log M \in \mathbb{N}$
General quantum state	Dirac ket with general state label	$ \psi\rangle$	$\in \mathbb{C}^M, n \geq \log M \in \mathbb{N}$
Qubit	Lowercase q (sans serif)	$\mathbf{q}$	Two-dimensional physical system
Qubit register	Uppercase roman (calligraphic)	$\mathcal{A}$	List of $m \leq n$ qubits $\mathbf{q}_i$
Quantum memory	Uppercase Q (calligraphic)	$\mathcal{Q}$	List of n qubits $\mathbf{q}_i$
Global quantum state	Dirac ket with quantum memory as label	$ \mathcal{Q}\rangle$	$\in \mathbb{C}^N$
Unitary transformation	Uppercase roman (sans serif)	$\mathbf{U}$	$\in \mathbb{C}^{M \times M}, n \geq \log M \in \mathbb{N}$
Apply unitary on register	Apply(unitary, register)	Apply($\mathbf{U} \rightarrow \mathcal{A}$)	Method (see Def. 4)

4 Defining Quantum Random Oracles

A classical oracle $\mathcal{O}_f$ is a black-box procedure that, given input x , returns $f(x)$ in such a way that no information about f other than that which can be inferred from its output is provided. A classical random oracle, then, is simply a classical oracle where the function f was chosen uniformly at random from the space of all functions from $\{0, 1\}^m$ to $\{0, 1\}^n$.¹⁰

However, the above operation is not unitary, as required for evolutions of quantum states (which is what we mean by “quantum algorithms”). Therefore, any quantum algorithm must instead be given access to *quantum-accessible* oracles, which on input the computational basis state $|x, y\rangle$, returns the unitary evolution of the state $|x, y \oplus f(x)\rangle$. In general (letting n and m be the length of the x and y registers, respectively), on input a quantum state $|\psi\rangle = \sum_{i \in [2^{n+m}]} \alpha_i |x_i, y_i\rangle$, we get:

$$\sum_{i \in [2^{n+m}]} \alpha_i |x_i, y_i\rangle \xrightarrow{\mathcal{O}_f} \sum_{i \in [2^{n+m}]} \alpha_i |x_i, y_i \oplus f(x_i)\rangle.$$

To see that this operation is unitary, it is enough to observe that it is *reversible*: Inputting the same state a second time yields the original state:

$$\sum_{i \in [2^{n+m}]} \alpha_i |x_i, y_i \oplus f(x_i)\rangle \xrightarrow{\mathcal{O}_f} \sum_{i \in [2^{n+m}]} \alpha_i |x_i, y_i \oplus f(x_i) \oplus f(x_i)\rangle = \sum_{i \in [2^{n+m}]} \alpha_i |x_i, y_i\rangle.$$

Sometimes it we will want to provide a quantum algorithm with access to a classical oracle.

Hans: Next.

5 Reprogramming QROs: The O2H Lemma and Friends

5.1 Recording ROs (or, Random Oracles are Good PRFs)

Hans: Copy-pasted from first intro draft. *Maybe* do the following later when we introduce the compressed random oracle technique? But definitely do not *start* here. (Game figs. in comments.) (Is O2H applicable to this though? Maybe we could provide proofs using both techniques?)

We start with the simplest ROM proof imaginable: showing that random oracles are good pseudo-random functions. Specifically, we will show that $\text{PRF}_k(x) = \text{H}(k||x)$ is indistinguishable from a random function, provided the key k is hard to guess.

We will do this by showing that if an adversary $\mathbf{A}$ can distinguish the output of PRF_k from random, then a reduction $\mathbf{B}$ can recover the key k . Crucially, the reduction does this by *recording* the queries of the adversary $\mathbf{A}$ to the random oracle $\mathbf{H}$.

¹⁰ Typically, random oracles follow hash functions in being defined from the space of all finite-length bit strings $\{0, 1\}^*$ to the space of n -length bitstrings $\{0, 1\}^n$; technically, this simply means that f is a collection of random functions: one per input length. Since quantum random oracles must operate by unitary means, we will explicitly restrict the domain of the random function to $\{0, 1\}^m$.

References

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A Select Exercise Solutions

Exercise 1. Let U be such that, for all states $|\psi\rangle$, $\|U|\psi\rangle\| = \||\psi\rangle\|$. Using the definition of the norm, this gives us that $|\langle U|\psi\rangle^\dagger U|\psi\rangle| = |\langle\psi|U^\dagger U|\psi\rangle| = |\langle\psi|\psi\rangle|$. Since this should hold for any $|\psi\rangle$, we must have that $U^\dagger U = \mathbb{1} = UU^\dagger$; the latter equality follows from the fact that, for all $|\varphi\rangle$ and $|\xi\rangle$, $|\langle\varphi|\xi\rangle| = |\langle\xi|\varphi\rangle|$. Thus, U satisfies the definition of an invertible matrix, with the inverse $U^\dagger$. Further, it is easy to see that the opposite also holds: any unitary matrix must be norm-preserving. Therefore, the set of norm-preserving operations on N -dimensional Hilbert space equal to the set of $N \times N$ unitary matrices.